

E-Commerce Cart Abandonment & Intent Classification Report

1. Project Overview

This project analyzes customer shopping behavior using over 411 million transactional clickstream logs to solve a critical financial bottleneck: an indiscriminate, automated discounting strategy.

- **The Business Problem (Revenue Bleed):** An analytics audit of a high-growth e-commerce platform revealed a massive cart abandonment rate driving a **\$2.17 billion revenue bleed**. However, the root cause of this financial drain was the company's "dumb discount system." In an attempt to recover carts, the platform automatically triggered a 10% discount pop-up and sent 24-hour reminder emails to every single user. This spray-and-pray approach caused a massive two-fold financial bleed:
 - *Margin Erosion (Setting Profit on Fire):* Unintentionally giving 10% discounts to "Safe Buyers"—users with high intent who were going to check out at full price anyway.
 - *Wasted Ad Spend (Plummeting ROAS):* Burning paid retargeting budget on "Window Shoppers"—users utilizing the cart as a wishlist with zero purchase intent.
- **The Business Objective:** The company possessed raw clickstream data but lacked a rule-based diagnostic engine to classify user intent. The objective of this project is to build an automated behavioral classification framework that segments users to protect profit margins, eliminate wasted ad spend, and surgically recover lost sales.
- **Core Business Questions Answered:**
 1. *The Baseline:* How much money is actually sitting in abandoned carts?
 2. *The Margin Bleed:* Who are the "Safe Buyers" we are accidentally giving discounts to?
 3. *The Noise:* Who are the "Window Shoppers" we should ignore completely?
 4. *The Rescue Target:* Who are the "Hesitant Buyers" that actually need the 10% pop-up to convert?

2. Dataset Summary

- **Total Rows:** 411,709,736
- **Time Period:** 7 Months (October 2019 to April 2020)
- **File Structure:** 7 compressed .csv.gz files processed across AWS and Google Cloud Platform (GCP)

- **Key Features:**
 - **User Identifiers:** user_id, user_session
 - **Event Details:** event_type (view, cart, purchase), event_time, price, category_code
 - **Behavioral Metrics:** time_to_cart_sec, unique_categories_viewed, views_after_last_cart
- **Data Scale:** Production-grade multi-category store clickstream logs requiring cloud-compute ELT ingestion.

3. Data Engineering & Transformation (dbt + SQL)

We began with data pipeline ingestion and schema cleaning using AWS Glue, Athena, BigQuery, and dbt:

- **Cloud Ingestion:** Loaded compressed raw files into AWS S3 buckets and Google Cloud Storage, utilizing AWS Glue Crawlers to establish a virtual database (analytics_db).
- **Price Sensitivity Baseline:** Used APPROX_PERCENTILE(price, 0.90) in SQL to dynamically calculate the 90th percentile price and average price independently for every product category.
- **Feature Engineering:**
 - Computed time_to_cart_sec by tracking the exact time elapsed between the first product view and the last cart addition .
 - Formatted raw seconds into executive-readable timestamps (0d:00h:00m:00s).
 - Created an overthinking ratio by segmenting total_views_up_to_last_cart against views_after_last_cart.
- **Intent Triage Engine:** Engineered a rule-based CASE statement to classify user sessions into four distinct behavioral buckets: Safe Buyer, Hesitate Buyer (Purchased), Hesitate Buyer (Abandoned), and Window Shopper.

4. Data Analysis using SQL (Business Transactions)

We performed structured SQL analysis in AWS Athena and BigQuery to answer the four core business questions:

1. Baseline Revenue Bleed - How much money is actually sitting in abandoned carts?

Metric	Value
Potential Revenue	\$3.89bn
Actual Revenue	\$1.73bn
Revenue Bleed	\$2.17bn

2. Margin Bleed - Who are the Safe Buyers we are accidentally giving discounts to?

Intent Segment	Sessions	Potential Revenue	Protected Margin (10%)
Safe Buyer	2.39M	\$833.76M	\$83.38M

3. Wasted Ad Spend - Who are the Window Shoppers we should ignore completely?

Intent Segment	Sessions	Abandoned Value	Actionable Status
Window Shopper	2.40M	\$841.40M	Non-Recoverable / Exclude from Ads

4. Rescue Target - Who are the Hesitant Buyers that actually need the 10% pop-up to convert?

Status	Sessions	Amount	Revenue Classification
Hesitate Buyer (Abandoned)	3.57M	\$1.33bn	Recoverable Revenue
Hesitate Buyer (Purchased)	2.30M	\$893.27M	Friction Revenue

5. ROI Simulation & Net Impact - What is the net financial return at a 4.00% conversion lift among Hesitant Buyers?

Financial Metric	Simulated Value
Gross Recovered Revenue	\$53.00M
Discount Expenditure (10%)	\$5.30M
Net New Revenue Generated	\$47.70M

6. Business Recommendations

- Suppress Safe Buyer Discounts – Configure checkout logic to block automated 10% pop-ups and emails for users browsing a single category who check out in under 3 minutes, protecting \$83.38M in profit margin.
- Target Hesitate Buyers with Exit Incentives – Trigger 10% discount pop-ups exclusively for high-intent shoppers displaying cart friction or exit intent to capture \$47.70M in net new revenue.
- Purge Window Shoppers from Paid Ads – Exclude users browsing 4+ categories over 10+ minutes from retargeting campaigns to eliminate ad budget waste on \$841.4M of non-recoverable carts.
- **Simplify the Payment Process (Reduce Checkout Friction)** – Our data shows that \$893.27M in revenue came from customers who eventually purchased our products, but experienced delays and hesitation at the checkout page. We should simplify the checkout flow and add "One-Click Payment" options (like Apple Pay, Google Pay, or saved cards) so customers can complete their purchases instantly without getting stuck or delayed.